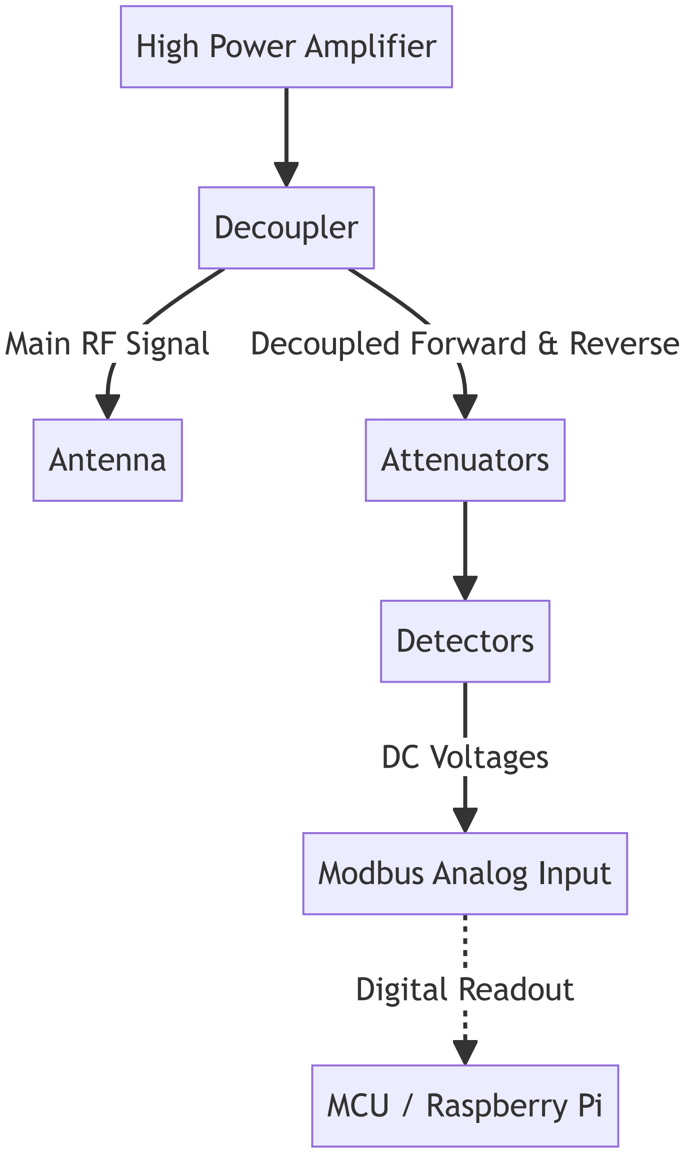
RF Power Measurement and VSWR Architecture

System Architecture

The system is designed to measure forward and reverse power by sampling the main RF path. The block diagram below illustrates the decoupled signal paths from the High Power Amplifier (HPA) to the MCU.



The **decoupler** samples the high-power RF signal and breaks out two attenuated RF signals: one for the forward power and one for the reverse (reflected) power.

It is standard practice to attenuate both these decoupled signals further. This ensures better matching with the detector's input range and minimizes standing waves in the decoupled circuit.

The **detector** is a device that takes the attenuated RF signal and outputs a DC voltage corresponding to the measured power of the signal. There is one detector for the decoupled forward power and one for the reverse power.

These DC voltages can then be read by the **Modbus analog input device** and communicated to the **MCU (Raspberry Pi)**, which executes calculations to figure out the VSWR.

Mathematical Calculations

The MCU performs the following calculations to derive VSWR from the measured voltages:

1. Voltage to Power Conversion

Read the voltage and convert it to power using the calibration curve for the specific detector module:

$$P_{\text{dBm}} = f_{\text{convert}}(V_{\text{measured}})$$

2. Logarithmic to Linear Power Conversion

Convert the power from logarithmic (dBm) to linear power (W):

$$P_{\text{Watts}} = \frac{10^{(P_{\text{dBm}}/10)}}{1000}$$

3. Reflection Coefficient (Γ)

Calculate the reflection coefficient:

$$\Gamma = \sqrt{\frac{P_{\text{reverse}}}{P_{\text{forward}}}}$$

4. Voltage Standing Wave Ratio (VSWR)

Calculate the VSWR using the reflection coefficient:

$$\text{VSWR} = \frac{1+\Gamma}{1-\Gamma}$$

Hardware Selection Guidelines

Picking a Decoupler

A dual-directional decoupler will break out two attenuated RF signals: one for the forward power and one for the reflected power.

- **Power and Frequency Ratings:** Make sure it is rated to handle the maximum output power of the HPA and operates within the correct frequency range. While broadband devices are available, using specific devices tailored to each HPA might yield better results.
- **Directivity:** Ensure the directivity is high enough. Higher directivity results in better separation of the forward and reverse power (minimal leakage), which directly translates to a higher accuracy VSWR measurement. This parameter should be matched to the detector module, which also has a maximum attainable accuracy.

Picking a Detector

An integrated circuit like the **LMH2110** from Texas Instruments is a strong candidate, but more research could be made here.

> **Note:** The LMH2110 is just a small microchip. There are supposedly EVALZ-style versions of the LMH2110 available that sit on a PCB with SMA input and DC output pins.

While evaluation boards are not typically designed for deployment in a professional setting, they are excellent for proof-of-concept. For a fully professional solution, a custom PCB based on the chosen detector chipset should be engineered. This custom board would provide a neat, finished solution containing:

- Two RF inputs (for forward and reverse power).
- Two sets of pins for the DC voltage outputs for direct connection to the Modbus analog input device.